VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY ▪ HCMUT ▪ FACULTY OF COMPUTER SCIENCE AND ENGINEERING

Food-101: Scratch CNN, Frozen Features, and Fine-Tuning

A comparative study of three adaptation regimes

Group choco (99 – PTQ)
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CO3133 – Deep Learning and Its Applications
Semester 261 ▪ Class CC01
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November 2026

The task: 101-way fine-grained food recognition

Why food is hard

- Huge intra-class variation: ingredients, plating, lighting, slicing.
- Subtle inter-class boundaries: *steak* vs. *filet mignon*, *tuna* vs. *beef tartare*.
- Non-rigid shapes and casual photography.
- Real-world label noise in the training set.

Research questions

1. Scratch CNN vs. pretrained ResNet-50?
2. Frozen features vs. full fine-tuning?
3. Which errors persist across models?

Metric: Top-1 accuracy and macro-F1 on 101 balanced classes.

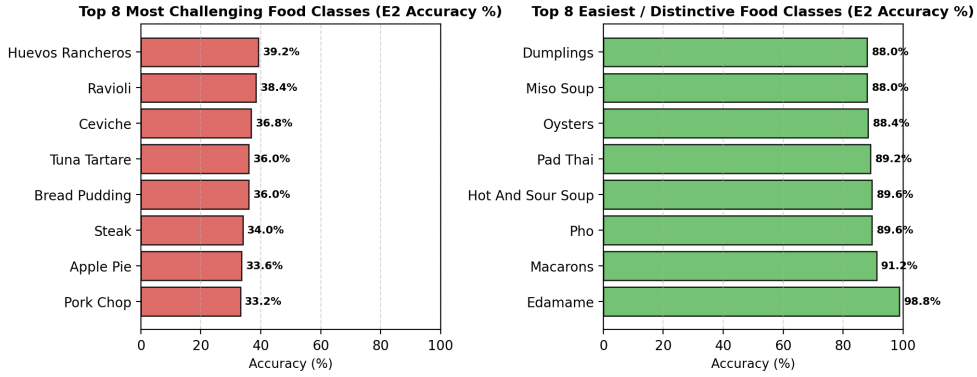
Dataset and leakage-free split

- **Food-101** (ETH Zürich): 101,000 images, 101 classes.
- Official: 750 train / 250 test per class, perfectly balanced.
- Our split: 90/10 stratified hold-out of the official training set, seed 42.
- Test set opened once per checkpoint.
- Eight culinary domains are used for analysis only – not an official label hierarchy.

Split	Images	/class
Train	68,175	675
Validation	7,575	75
Official test	25,250	250
Total	101,000	1,000

101 classes $\times$ 250 test images $=$ 25,250 final predictions.

Not all classes are equally hard



E2 accuracy ranges from **33.2%** (*pork chop*) to **98.8%** (*edamame*). Continuous textures and ambiguous cuts dominate the failures.

Three adaptation regimes

	E1: Simple CNN	**E2: ResNet-50 frozen**	**E3: ResNet-50 fine-tuned**
Initialisation	random	ImageNet-1k	ImageNet-1k
Trainable params	414,853	**206,949**	23,714,981
Frozen backbone	–	23,508,032	none
Optimiser	AdamW	Adam	AdamW
Learning rate	10^{-3}	10^{-3}	10^{-4}
Batch size	64	64	32
Epochs	20	15	15
GPU	T4	T4	P100

- E1 is a scratch reference; it differs from E2 in architecture and initialisation.
- E3 changes *both* unfreezing and the optimisation protocol.
- A0 (control): frozen backbone, E3 protocol (AdamW, 10^{-4}, batch 32).

What the comparison can and cannot claim

It is a protocol comparison

- E1 $\rightarrow$ E2: architecture *and* initialisation change.
- E2 $\rightarrow$ E3: unfreezing *plus* Adam$\rightarrow$AdamW, $10^{-3} \rightarrow 10^{-4}$, batch 64$\rightarrow$32, T4$\rightarrow$P100.
- The $+19.47$ points is therefore descriptive, not a clean causal estimate of unfreezing.
- A0 keeps the protocol fixed; in the two recorded runs the gap to E3 is **+24.46 points** (GPU and LR schedule not fully controlled).

Matched control: A0

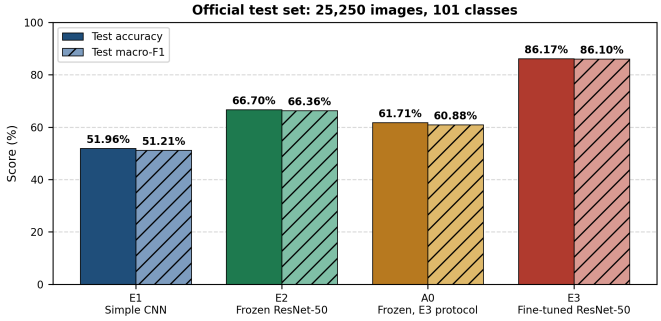
`ablation_frozen_e3proto.yaml`

Frozen backbone, E3 protocol.
61.71% / **60.88%** macro-F1

vs. **86.17%** / **86.10%** fine-tuned

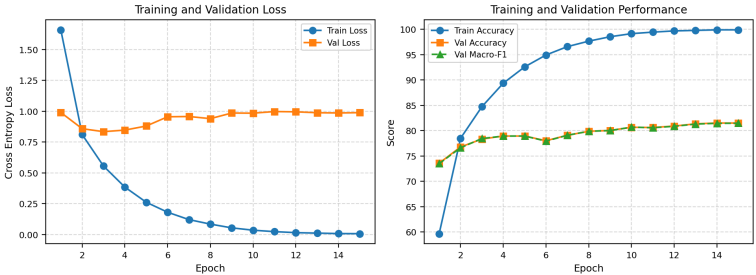
The protocol switch coincides with a 4.99-point drop for the frozen model.

Results on the official test set



E2 over E1: **+14.74 points** with *half* the trainable parameters.
A0 (frozen, E3 protocol): 61.71% / 60.88% – the protocol switch coincides with a 4.99-point drop.
E3: **86.17% accuracy**, **86.10% macro-F1**, loss 1.29 → 0.65; in the two recorded runs, **+24.46 points** over A0.

Training dynamics



- E1 stays close to its validation curve (train 43.6%, val 47.2%): capacity-limited, not overfitting.
- E3 fits the training set (99.9%) while validation plateaus at 81.5% after epoch 9 – the clearest overfitting signal.
- Motivation for CutMix, MixUp, and label smoothing.

Error analysis: the same semantic confusions persist

Rank	E1	E2	E3
1	*steak* → *filet mignon* (42)	*steak* → *filet mignon* (47)	*filet mignon* → *steak* (36)
2	*cheesecake* → *panna cotta* (40)	*pulled pork sandwich* → *hamburger* (32)	*steak* → *filet mignon* (32)
3	*samosa* → *spring rolls* (37)	*filet mignon* → *steak* (29)	*tuna tartare* → *beef tartare* (25)
4	*chocolate mousse* → *chocolate cake* (33)	*pork chop* → *filet mignon* (29)	*chocolate cake* → *chocolate mousse* (21)
5	*panna cotta* → *cheesecake* (28)	*chocolate cake* → *chocolate mousse* (24)	*steak* → *pork chop* (21)

- One-directional pairs from each run's `test_metrics.json`; mutual directions are separate entries.
- Beef cuts, raw-protein molds, and chocolate desserts survive fine-tuning.
- Cures: higher input resolution, hard-negative mining, contextual cues (bone, cut thickness).

Qualitative: failures are semantic neighbours



Top row: correct ▪ Bottom row: Ground truth → Prediction (E3)

Correct (top): red velvet cake, cup cakes, nachos, onion rings.

Incorrect (bottom):

- huevos rancheros → nachos
- tuna tartare → beef tartare
- lobster bisque → foie gras
- filet mignon → steak

Seeded shuffle, one image per true class.

Compute budget and practical trade-off

Run	GPU	Epochs	Batch	Train time	Trainable
E1	Tesla T4	20	64	162.52 min	414,853
E2	Tesla T4	15	64	**149.82 min**	**206,949**
A0	Colab (unlogged)	15	32	108.17 min	**206,949**
E3	Tesla P100	15	32	269.35 min	23,714,981

- Timing is wall-clock on different GPUs: *not* hardware-matched.
- Frozen backbone: 66.7% in $\sim$2.5 h on a T4 – strong low-cost baseline.
- Fine-tuning: 86.2% in $\sim$4.5 h on a P100 – best accuracy per submission.
- Inference is identical for E2 and E3: **221 vs. 223 img/s**, 530 MB peak VRAM (T4, fp32, batch 32). Unfreezing costs training time only.

Limitations

- **Residual protocol differences**: A0 controls the optimiser, LR, and batch size, but it ran on an unlogged Colab GPU and E2 differs from A0 in three settings.
- **Single seed** (42): no mean $\pm$ std across runs.
- **Overfitting**: 99.88% train vs. 81.50% validation in E3.
- **Fixed** 224^2 **input**: discards fine texture cues.
- **Partial confusion evidence**: full E1/E3 matrices were not exported; eight qualitative examples only.
- **Single-session inference measurement**: one fp32 T4 run, no TensorRT or AMP.
- **Single backbone family**: ResNet-50 only.

Conclusion

- A frozen ImageNet ResNet-50 beats a scratch CNN by **14.74 points** while training **fewer** parameters.
- Full fine-tuning reaches **86.17% accuracy / 86.10% macro-F1** on the official test set.
- In the two recorded runs, E3 exceeds A0 by **24.46 points**; GPU and LR schedule are not fully controlled, so the gap is descriptive. The protocol switch coincides with a 4.99-point drop for the frozen model.
- Remaining errors are semantic, not random: beef cuts, tartares, chocolate desserts.
- Next steps: CutMix/MixUp, 384^2 input, ConvNeXt/ViT under the same protocol.

Reproducibility

Code, configs, checkpoint contract: `assignment2/` ▪ `CHECKPOINTS.md` ▪ AI use disclosed in `AI_USAGE.md`.

Thank you

Questions?